Software Requirements Specification for Industrial Plant Designer: subtitle describing software

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Revision History

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April 3rd 2024	Madhu Sivapragasam	2.0	Removed unneeded requirement for rev 1

1 Purpose of the Project

1.1 User Business

Our sponsor, Professor Vladimir Mahelec is looking to create a system that can simulate a factory plant's behaviour depending on a given set of inputs. Currently, Professor Mahelec and his team have developed a model that takes input data from a set of Excel tables and then processes the data before sending output data back. This input data represents different components of the factory, variable data within the components, connections between components, etc. Due to the complexity of the input data, various Excel tables are needed to store and organize the varying inputs properly. Unfortunately, creating and updating data in these Excel tables can be a complex and difficult task due to the complexity of the workbook. Therefore, users of the system need to have familiarity with their table conventions and that can be difficult if this service is used by external clients. Similarly updating data in the table can be time-consuming since each components need to have new data added and new connections defined which can be a slow process.

To solve this, our sponsor is requesting us to build a user interface that will allow a user to easily input the factory plant data. This would include a schema drawing tool that will allow a user to create a schema along with tooling to input any other data. While this service is initially intended for hydrogen plants, it should be customizable so that all types of factories can use the service. Eventually, Professor Mahelec intends to sell this service to other vendors and plant operators so that they can use it themselves to visualize and simulate factory plant data.

1.2 Goals of the Project

- Develop a GUI that allows a user to create a factory plant schema, the data from this schema can then be sent to Professor Mahelec's model which will then produce the output which should also be displayed in the GUI
- Develop other UI tooling that can accept inputs that may not be available in a factory schema GUI such that all other input data can also be successfully entered by the user
- Develop a backend service and db layer that can the data from the UI inputs and send it to the processing model and then take the response from that model and send it back to the user

- Develop a version control system such that a user can make a plant schema, save it and then recover it at a later date
- Create documentation that allows for easy handover of the project after the project is complete so that our sponsor can take the project and use it easily themselves
- Accomplish all of Professor Mahelec's requirements that he sets for the service so that he is satisfied with the end result of the project

2 Stakeholders

2.1 Client

The client for this project is Dr. Vladimir Mahalec, who will serve as the primary point of contact throughout the development process.

2.2 Customer

The target customers for this product are engineers across various disciplines who need to model industrial plants using nodes and edges to represent machines and material streams. These engineers will also use the software to run simulations of plant processes and obtain results for further analysis.

2.3 Society

The impact of this product on society can be understood through several perspectives:

- **Efficiency in Industrial Operations:** By enabling engineers to model and simulate industrial plants, this software can help optimize plant efficiency, reduce resource consumption, and minimize waste. Improved operational efficiency can lower production costs and reduce the environmental footprint, benefiting both industry and society.
- **Advancement in Engineering Practices:** This product serves as a valuable tool for practicing engineers and students alike, allowing them to visualize complex systems and test different scenarios before real-world implementation. This can lead to more innovative designs, improved safety standards, and better overall outcomes in industrial engineering.

- **Educational Contribution:** Since the software will be used by Dr. Mahalec and his students, it contributes to advancing education and research in fields such as chemical and mechanical engineering. As a result, it supports the development of future engineers, enhancing their skills and preparedness for real-world challenges.

- **Sustainability and Environmental Impact:** By simulating industrial processes with a focus on optimizing energy use and reducing emissions, the software can aid industries in adopting more sustainable practices. This can have broader societal benefits by supporting efforts to combat climate change and promote environmentally responsible practices.

In summary, this product has the potential to positively affect society by improving industrial efficiency, advancing engineering education and practices, and contributing to sustainability efforts.

2.4 Other Stakeholders

Other relevant stakeholders include:

- Investors: Individuals or companies who fund the construction and operation of industrial plants. They will hire engineers who may use this product to model and optimize production processes.

- Supervisors and Academics: Faculty members, such as Dr. Mahalec and his post-graduate students, who will provide guidance during the development and evaluation of the product.

2.5 Hands-On Users of the Product

The hands-on users of this product include:

- Supervisors and Researchers: Dr. Mahalec and his post-graduate students, who will be involved in testing, providing feedback, and ensuring the product meets the necessary standards.

- Engineers: The primary end users, who will use the product to model, simulate, and analyze the operations of industrial plants.

2.6 Personas

- **Process Engineer (Jane Smith)**

  Background: Jane is a process engineer with 5 years of experience in the energy sector, focusing on chemical processes like hydrogen production and refining. She holds a Master's degree in Chemical Engineering.

  Goals: Jane uses the software to design detailed models of industrial plants, focusing on optimizing energy efficiency. She frequently runs simulations to test different operational scenarios and identify bottlenecks in the system.

  Challenges: Jane is pressed for time and values user-friendly tools that allow her to visualize plant models quickly. She needs precise simulation results to support decision-making and regulatory compliance.

- **Mechanical Engineer (Sam Lee)**

  Background: Sam is a mechanical engineer specializing in designing and maintaining industrial equipment. He has a Bachelor's degree in Mechanical Engineering and has worked in the field for 8 years.

  Goals: Sam uses the software to map out the physical layout of machinery and ensure that the plant model accurately reflects the equipment's configuration and operation. He's focused on improving the reliability of machinery and minimizing downtime.

  Challenges: Sam requires the software to integrate mechanical constraints, such as pressure and flow rates, to ensure the safety and efficiency of the modeled systems.

- **Graduate Researcher (Maria Gomez)**

  Background: Maria is a graduate student working under Dr. Mahalec, conducting research on advanced simulation techniques for industrial processes. She is pursuing a PhD in Chemical Engineering.

  Goals: Maria uses the software as part of her research to simulate theoretical plant models. She experiments with new configurations and operating conditions to evaluate their potential for energy savings and emissions reduction.

  Challenges: Maria needs advanced features that allow her to customize simulations and compare results across multiple scenarios. She values flexibility and the ability to export data for further analysis in other tools.

2.7 Priorities Assigned to Users

The highest priority will be given to users who have a significant stake in the success of the product. This includes Dr. Vladimir Mahalec, as the client and primary stakeholder, and his post-graduate students, who will rely on the software for research and academic purposes. Their feedback and requirements will be critical in shaping the development process and ensuring that the final product meets the intended goals.

2.8 User Participation

During the development phase, most user participation will come from the project developers and supervisors. The supervisors, particularly Dr. Mahalec and his post-graduate students, will play a key role in providing feedback and testing the product. Their input is crucial, as they will be the primary users during the initial stages and will help ensure that the software aligns with its intended functionality and usability goals.

2.9 Maintenance Users and Service Technicians

The development team will be responsible for maintaining the software until April 2025. After this period, Dr. Vladimir Mahalec will acquire the rights to the product, assuming responsibility for its ongoing maintenance and support. Any future updates, bug fixes, or enhancements will fall under his purview or that of any designated technical team.

3 Mandated Constraints

3.1 Solution Constraints

3.1.1 Dual Database System Implementation

Description: The software must implement a dual database system, consisting of separate user and system databases. The user database will store user-specific data like customized plant models, parameters, and configurations. The system database will store standard industry data, default equipment parameters, and operational constraints.

Rationale: The separation of user and system data has been explicitly requested

by the supervisor to ensure data integrity, security, and ease of maintenance. This dual database setup allows flexible customization of plant models while maintaining the integrity of standard system data.

Fit Criterion:

- The system must use two distinct databases: one for system data and one for user data.
- User modifications must only affect the user database without altering the system database.
- The software shall prevent any write operations to the system database from the user interface.

3.1.2 Web-Based Canvas Interface with Nodes and Arcs

Description: The software must provide a web-based graphical user interface (GUI) that runs in a browser, featuring a canvas where users can create plant models using draggable nodes (representing equipment) and connect them with arcs (representing material or energy streams). The supervisor strongly suggested using React Flow which is open-source nodes and arcs library written in typescript.

Rationale: To improve usability and move away from manual spreadsheet entries, the supervisor **requires** a visual, interactive interface accessible via web browsers. This enables easier access and collaboration and provides an intuitive way for users to build and visualize complex plant models

Fit Criterion:

- The GUI must be accessible through standard web browsers (e.g., Chrome, Firefox, Edge).
- The suggestion of React Flow library enforces that we shall use Typescript/Javascript for the client facing code (frontend).
- Users shall be able to select, drag, and drop equipment icons onto a canvas to create nodes.
- Users must be able to draw arcs connecting nodes to represent streams.
- The canvas shall support zooming, panning, and hierarchical views for subnetworks.

3.1.3 Integration with Existing Python Mathematical Models

Description: The software must interface seamlessly with the existing Python scripts developed by Dr. Mahalec's team for mathematical modeling and simulation. This integration may be done via an API or direct database connections to provide necessary parameter data for computations.

Rationale: The mathematical computations and optimizations are crucial and have already been developed in Python. Integrating the new software with these existing scripts will ensure continuity, leveraging prior work, and avoid redevelopment of complex algorithms and is mandated by his team.

Fit Criterion:

- The software must provide an interface (API or database connection) that allows Python scripts to access necessary parameter data.
- Data exchanged between the software and Python scripts shall be in a compatible format.

3.1.4 Table Format Interface for Data Editing

Description: The software must provide a table format interface for editing data, irrespective of the database system chosen. This interface shall allow users to view, add, modify, and delete data entries in a structured tabular format, facilitating easy data management alongside the graphical modeling tools.

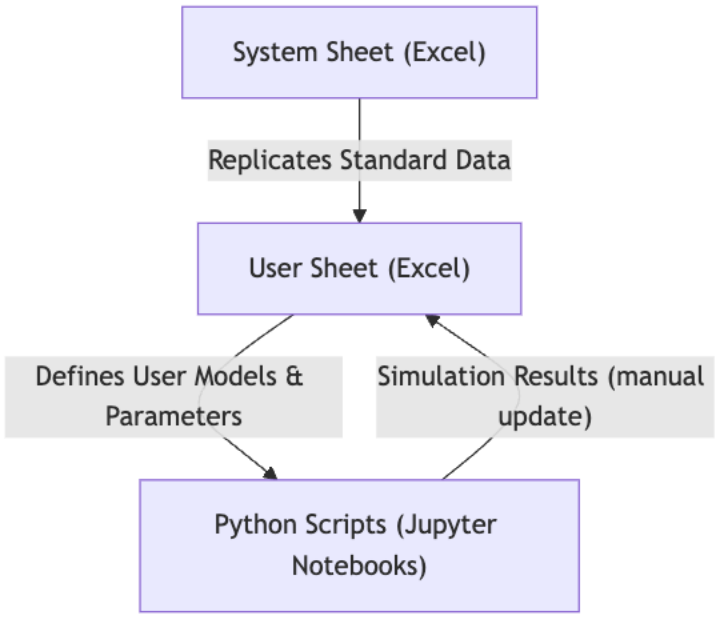
Rationale: Dr. Mahalec emphasized the need for a table format interface to ensure users can manage data efficiently, not relying solely on the graphical canvas.

Fit Criterion:

- The table interface must present data in a clear and organized manner, mirroring the underlying database structure.
- Users must be able to perform CRUD (Create, Read, Update, Delete) operations within the table interface with immediate reflection in the database.
- Changes made through the table interface must automatically update the graphical model on the canvas and vice versa, ensuring data consistency across both interfaces.

3.2 Implementation Environment of the Current System

Description: The current system used by Dr. Mahalec's team relies on Microsoft Excel for data management and Python scripts for mathematical modelling and simulation. It includes two main Excel sheets: **User Sheet and System Sheet** —containing over 30 workbooks that define hundreds of chemical parameters needed for plant simulations.



Important components are:

- **User Sheet:**
  - **Content:** Contains user-specific data, including customized plant models, parameters, and configurations.
  - **Structure:** Over 30 workbooks detailing aspects like node parameters, stream properties, and operational constraints.
- **System Sheet:**
  - **Content:** Stores standardized industry data, default equipment parameters, and foundational models.

  - **Structure:** Over 30 workbooks define templates for node models, ports, variables, and thermodynamic properties.

- **Python Scripts:**
  - **Environment:** Executed in Jupyter Notebooks.
  - **Functionality:** Perform modeling and simulations based on data from Excel sheets.
  - **Libraries used:** Python 3.8 with libraries such as NumPy, Pandas, and SciPy.

- **Data Handling:**
  - **Manual Processes:** Data is manually entered and updated in Excel. Parameters are manually referenced or exported for Python.
  - **Simulation Workflow:** After data entry, Python scripts are manually run to perform simulations. Results are reviewed and documented externally.

3.3 Partner or Collaborative Applications

Existing Python-Based Mathematical Models:

Description: Python scripts developed in Jupyter Notebooks by Dr. Mahalec's team to perform complex mathematical modelling and simulations based on plant data.

Software Interfaces: The new software will interface with these scripts through a well-defined API or direct database connections, allowing seamless access to parameter data and the input/output of simulation results.

3.4 Off-the-Shelf Software

Product Name	Usage	Rationale
ReactFlow	ReactFlow will be used as a React-based library to manage the creation and manipulation of nodes and arcs within the graphical user interface (GUI).	Dr. Mahalec advised using ReactFlow due to its features for handling complex node and arc interactions. It integrates with React, making development smoother, and has an active community that provides continuous support and updates.
dbgate	dbgate will be used as the database management tool to provide users with a table-like interface for managing and editing data within the databases.	Dr. Mahalec recommended dbgate for its versatility in handling both SQL and NoSQL database systems. It offers a user-friendly interface that simplifies data management tasks, allowing users to perform operations without needing specialized database training, making it an good choice for the project.

3.5 Anticipated Workplace Environment

The software will be used by individual users in a standard office setting, primarily on desktop or laptop computers. Users will access the web-based GUI through widely-used web browsers like Chrome, Firefox, or Edge. Each user may manage multiple plant models simultaneously, meaning the software must run smoothly on typical office hardware without requiring any specialized equipment. Since the office environment will have consistent internet access, this will support the web-based interface and ensure smooth integration with any necessary local systems.

3.6 Schedule Constraints

As stated in the SFWRENG 4G06 course outline, the schedule constraints are as follows:

Task	**Date**
Requirements Document Revision 0	October 9
Hazard Analysis 0	October 23
V&V Plan Revision 0	November 1
Proof of Concept Demonstration	November 11 to November 22
Design Document Revision 0	January 15
Revision 0 Demonstration	February 3 to February 14
V&V Report Revision 0	March 7
Final Demonstration (Revision 1)	March 24 to March 30
EXPO Demonstration	April TBD
Final Documentation (Revision 1)	April 2

Table 1: Project Timeline

3.7 Budget Constraints

This project will follow a zero-budget framework, relying entirely on free-tier and open-source software to minimize costs. All software components will run locally, avoiding any additional expenses apart from the purchase of ReactFlow Pro, which is required to access advanced node and arc examples as advised by Dr. Mahalec. If any purchases are necessary to meet the capstone deliverable requirements, the costs will be covered by the team members, with a total budget limit of $750. Once the project is handed over to Dr. Mahalec, it will revert to operating under the zero-budget framework.

3.8 Enterprise Constraints

The software must operate entirely on local machines to comply with the enterprise's data security and operational independence policies. This prohibits the use of exter-

nal paid services and ensures that all data processing and storage are done within the organization's internal infrastructure. Following these constraints guarantees the software maintains data integrity and aligns with enterprise standards for security and software usage.

Although the project will mostly use free-tier and open-source libraries, certain commercial components like ReactFlow Pro will be included to meet specific functional requirements. All components, whether commercial or open-source, must meet the organization's compatibility and security standards, ensuring seamless integration within the local environment. This approach balances cost-efficiency with the use of specialized tools necessary for the project's success.

4 Naming Conventions and Terminology

4.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

- UI: User Interface, refers to an interface that users can use to enter data inputs
- GUI: Graphical User Interface, refers to a user interface that includes graphics/images to reduce data complexity for theuser
- API: Application Programming Inteface: An abstracted programming layer that allows a user to interface with more complex systems without needing to have an understanding of the internals of that system
- : Node: Refers to a single machine in a factory
- : Edge: Refers to a connection between factory machines
- : Graph: Refers to a display that shows a series of nodes as shapes and a series of edges as arrows/lines
- Factory plant/plant: Refers to a factory building that is responsible for processing some sort of input into some sort of outputs through a series of nodes (factory machines) and edges (factory connections)
- Plant schema/schema: Refers to a graph shaped design for a factory plant via a series of nodes and edges on canvas

5 Relevant Facts And Assumptions

Relevant Facts

- The Python scripts simulate plant operations such as material flow and energy usage but currently require manual inputs and command-line execution.
- The final software, including code and documentation, will be owned by the client (Professor Mahalec) upon project completion, ensuring future use and modifications align with research objectives.
- Security measures, including NDAs and controlled data access, are in place to protect sensitive information and intellectual property throughout the project.

5.1 Business Rules

- All simulation configurations, results, and user-generated data must be stored in a secure and accessible manner, with integrated services for future retrieval and analysis.
- The software should integrate seamlessly with existing Python models for plant simulations, ensuring accurate and efficient execution of simulations based on user inputs.
- The users want to use Business Intelligence tools outside of the application to analyze the output.

5.2 Assumptions

- The existing Python models are functional and will be provided by Dr. Mahalec's team for integration into the new software solution.
- The project team has access to the necessary hardware and software resources required for development and testing, including the Python models.
- The project scope, as defined by the team and Professor Mahalec, will not change significantly during the course of development, barring any unforeseen technical challenges or new requirements from the client.
- It is assumed that Professor Mahalec will provide regular feedback and technical guidance to ensure the project stays aligned with research and educational goals.

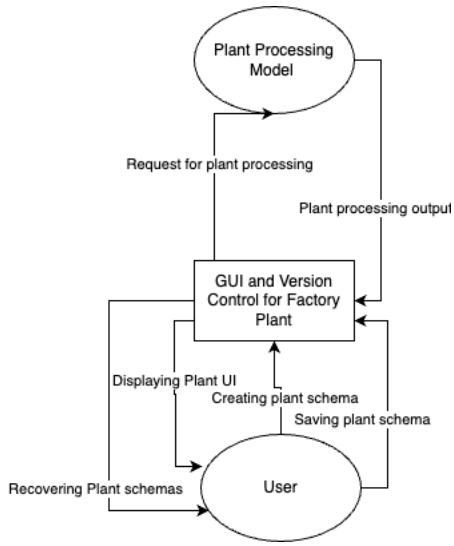
- The software will primarily be used in an academic or research setting, and its potential deployment in an actual industrial environment would require further refinement and validation.
- The project will be developed using open-source tools and libraries, except where proprietary IP (e.g., the Python models) is used. Licensing for third-party tools will be managed in compliance with McMaster's policies and relevant copyright laws.

6 The Scope of the Work

6.1 The Current Situation

Currently, there exists a Python model that takes in data from a series of Excel tables, uses that data to simulate a plant and then outputs the results from that simulation. There is no UI or standardized method of entering the data nor is there a formal version control system that allows multiple users to save plant data and then recover it to continue working on it. Our solution produce a UI that would allow users to interface with the model without having the manually enter data while also providing a version control system that enables the user to save and recover work when needed.

6.2 The Context of the Work



In the above context diagram, we see the surrounding context of the system we are building. Our system is displayed by the square box in the center labeled GUI and version control for the factory plant with the two other nodes in the diagram representing external entities that our system needs to interact with. The arrows show the inputs/outputs of these different entities with the system. We can see that a user has various interactions with the system such as updating the schemas, saving schemas, and restoring previous schemas. The system in turn provides the user with a UI that allows them to visualize the plant as a response to these inputs. Similarly, our system interacts with an external Python model by sending it data and then receiving outputs from that data.

6.3 Work Partitioning

Event name	Input and Output	Summary of BUC
Create factory node/edge	Create plant schema (in)	A new node or edge is added by the user and displayed in the schema canvas
Delete factory node/edge	Create plant schema (in)	The user deletes a given node/edge from the schema canvas
Move node/edge	Create plant schema (in)	The user moves the position of a node or edge in the schema
Save schema	Saving plant schema (in)	The user clicks to save a copy of the plant schema data
Restore schema	Recovering plant schema (in)	The user clicks to restore/recover an existing saved schema
Display schema	Displaying plant UI (out)	The system displays the current canvas to the user through the GUI
Send factory data	Request for plant processing (out)	The system makes a request to the plant processing model containing the data of the factory inputs in order to get the factory outputs
Receive factory outputs	Plant processing output (in)	The system receives the plant processing output data

6.4 Specifying a Business Use Case (BUC)

The specification for each business use case is detailed in the above table however a short summary below will provide more info. The business use cases can be summarized into four main types, user inputs, graphical output to user, request to model and response from model. The user inputs are when the user updates the schema, saves the schema and restores the schema and these are carried out via button press

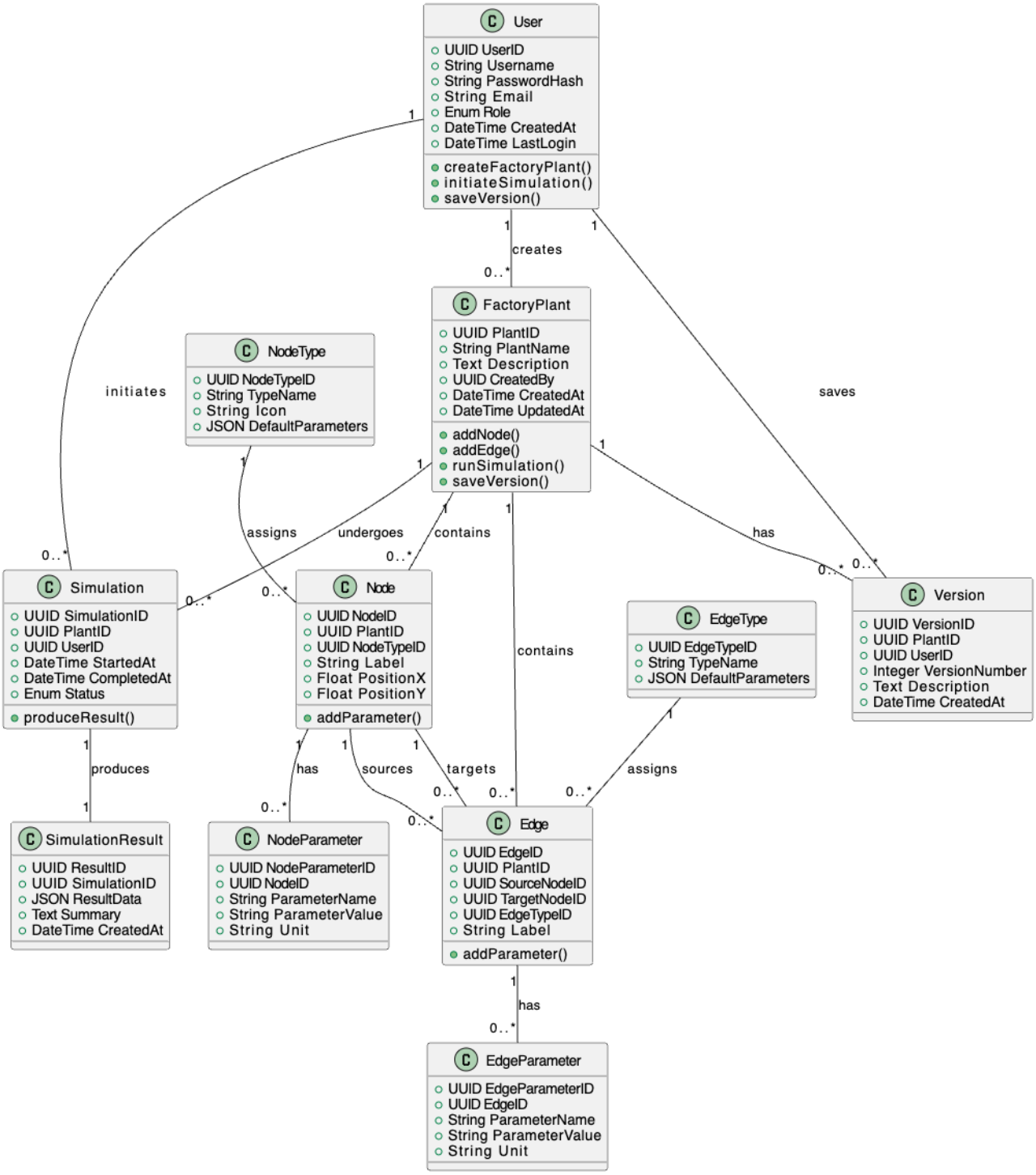
events in the system. The graphical output of the system is just the schema diagram displayed on a webpage for the user. The request to the model is an HTTP request carried out by the backend of our system to the model which lives in its own server. This request carries data in the form of an HTTP request body that contains the factory parameters. Finally, the response from the model is in the form of an HTTP response where in the response body we can find the output data of the factory.

7 Business Data Model and Data Dictionary

7.1 Business Data Model

In the Class Diagram below User entity is main, representing everyone interacting with the system, from engineers to supervisors as mentioned in the stakeholders section. Each user can create multiple FactoryPlants, which represents different plant setups. Within each FactoryPlant, there are multiple Nodes and Edges. The Nodes represents different machines like reactors and pumps, each with its own set of parameters such as temperature and pressure. The Edges class, show the streams connecting these nodes, showcasing the flow of materials and info.

Additionally, there is NodeType and EdgeType classes that classifies the different kind of Nodes and Edges available. For example, a NodeType defines a specific type of pump, while an EdgeType could specify whether a stream is gas or liquid. Each FactoryPlant/Map can undergo multiple Simulations, and each Simulation generates a SimulationResult, showing the results of the Python Script output ran by Dr. Mahalec's team. To keep track of changes and different versions of a FactoryPlant for different time period, there is also a Version class that allows users to look at different time period and change parameters for different period.


User
UUID UserID
String Username
String PasswordHash
String Email
Enum Role
DateTime CreatedAt
DateTime LastLogin
createFactoryPlant()
initiateSimulation()
saveVersion()
creates
FactoryPlant
UUID PlantID
String PlantName
Text Description
UUID CreatedBy
DateTime CreatedAt
DateTime UpdatedAt
addNode()
addEdge()
runSimulation()
saveVersion()
NodeType
UUID NodeTypeID
String TypeName
String Icon
JSON DefaultParameters
initiates
saves
assigns
undergoes
contains
has
Simulation
UUID SimulationID
UUID PlantID
UUID UserID
DateTime StartedAt
DateTime CompletedAt
Enum Status
produceResult()
Node
UUID NodeID
UUID PlantID
UUID NodeTypeID
String Label
Float PositionX
Float PositionY
addParameter()
EdgeType
UUID EdgeTypeID
String TypeName
JSON DefaultParameters
Version
UUID VersionID
UUID PlantID
UUID UserID
Integer VersionNumber
Text Description
DateTime CreatedAt
produces
has
sources
targets
assigns
SimulationResult
UUID ResultID
UUID SimulationID
JSON ResultData
Text Summary
DateTime CreatedAt
NodeParameter
UUID NodeParameterID
UUID NodeID
String ParameterName
String ParameterValue
String Unit
Edge
UUID EdgeID
UUID PlantID
UUID SourceNodeID
UUID TargetNodeID
UUID EdgeTypeID
String Label
addParameter()
has
EdgeParameter
UUID EdgeParameterID
UUID EdgeID
String ParameterName
String ParameterValue
String Unit

7.2 Data Dictionary

Class Name	Attribute Name	Data Type	Description
User	UserID	UUID	Unique ID for each user
	Username	String	The user's login name
	PasswordHash	String	Hashed password for auth
	Email	String	User email address
	Role	Enum	User roles eg: admin, user
	CreatedAt	DateTime	Timestamp when account created
	LastLogin	DateTime	Timestamp last user login
FactoryPlant	PlantID	UUID	Unique id for factory plant
	PlantName	String	Name of the factory plant
	Description	Text	Description of factory plant
	CreatedBy	UUID (FK to User)	References the User who created the plant
	CreatedAt	DateTime	Timestamp of when the plant was created
	UpdatedAt	DateTime	Timestamp of when the plant was last updated
Node	NodeID	UUID	Unique id for each node
	PlantID	UUID (FK to FactoryPlant)	Reference to the parent factory plant
	NodeTypeID	UUID (FK to NodeType)	Reference to the type of node
	Label	String	Display label for the node
	PositionX	Float	X-coordinate position on the canvas
	PositionY	Float	Y-coordinate position on the canvas
Edge (Stream)	EdgeID	UUID	Unique id for each edge
	PlantID	UUID (FK to FactoryPlant)	Reference to the parent factory plant
	SourceNodeID	UUID (FK to Node)	Reference to the source node

Class Name	Attribute Name	Data Type	Description
	TargetNodeID	UUID (FK to Node)	Reference to the target node
	EdgeTypeID	UUID (FK to EdgeType)	Reference to the type of edge
	Label	String	Display label for the edge
NodeType	NodeTypeID	UUID	Unique id for each node type
	TypeName	String	Name of the node type eg: Reactor, Pump
	Icon	String	Icon representing the node (svg)
	DefaultParameters	JSON	Default parameters for this node type
EdgeType	EdgeTypeID	UUID	Unique id for each edge type
	TypeName	String	Name of the edge type eg: Gas Stream, Liquid Stream, etc
	DefaultParameters	JSON	Default parameters for this edge type
NodeParameter	NodeParameterID	UUID	Unique ID for each node parameter
	NodeID	UUID (FK to Node)	Reference to the node
	ParameterName	String	Name of the parameter eg: Temperature, Pressure, etc
	ParameterValue	String/Float	Value of the parameter.
	Unit	String	Unit of measurement eg: Celsius, Bar, etc
EdgeParameter	EdgeParameterID	UUID	Unique ID for each edge parameter
	EdgeID	UUID (FK to Edge)	Reference to the edge
	ParameterName	String	Name of the parameter eg: Flow Rate, Composition, etc
	ParameterValue	String/Float	Value of the parameter
	Unit	String	Unit of measurement eg: Liters per Minute
Simulation	SimulationID	UUID	Unique ID for each simulation run

Class Name	Attribute Name	Data Type	Description
	PlantID	UUID (FK to FactoryPlant)	Reference to the factory plant being simulated
	UserID	UUID (FK to User)	Reference to the user who initiated the simulation
	StartedAt	DateTime	Timestamp when the simulation started
	CompletedAt	DateTime	Timestamp when the simulation completed
	Status	Enum	Current status of the simulation eg: Pending, Running, Completed, Failed
SimulationResult	ResultID	UUID	Unique ID for each simulation result
	SimulationID	UUID (FK to Simulation)	Reference to the simulation
	ResultData	JSON/Blob	Data output from the simulation
	Summary	Text	Summary of the simulation results.
	CreatedAt	DateTime	Timestamp when the result was created
Version	VersionID	UUID	Unique ID for each version
	PlantID	UUID (FK to FactoryPlant)	Reference to the factory plant associated with the version
	UserID	UUID (FK to User)	Reference to the user who saved the version
	VersionNumber	Integer	Version number based on time period
	Description	Text	Description
	CreatedAt	DateTime	Timestamp when the version was saved

8 The Scope of the Product

8.1 Product Boundary

The scope of the project includes the development of a software solution for modeling and simulating industrial plant operations. The software will consist of a graphical user interface (GUI) for building plant models, integrating with existing Python scripts through a backend server for simulating plant operations, and a database system for storing user configurations and simulation results. The product is being developed by Team 18 at McMaster University under the supervision of Professor Vladimir Mahalec. The goal is to improve the efficiency and accuracy of plant modeling while providing a user-friendly interface and robust backend capabilities.

The software will include the features and functionalities as defined in the requirements sections 9-17 of this document:

8.2 Product Use Case Table

The primary use cases of the software are outlined below:

Use Case ID	Description
UC-1	**Create Plant Model**: Users can create a visual model of an industrial plant using the drag-and-drop GUI, adding nodes and edges to represent various processes and their interactions.
UC-2	**Run Plant Simulation**: Users can input data and run a simulation using the existing Python models. The simulation results will be displayed in the GUI.
UC-3	**Save and Load Plant Models**: Users can save their plant models to a local or remote database, and load existing models for modification or review.
UC-4	**View Simulation Results**: Simulation outputs, such as production rates, efficiency metrics, and potential bottlenecks, will be presented to users through the GUI.
UC-5	**Export Data**: Users can export simulation results and configuration data to external formats such as CSV or Excel for further analysis.
UC-6	**Manage User Accounts**: The software will support multi-user authentication, allowing different users to create and access their models securely.

Table 3: Product Use Case Table

8.3 Individual Product Use Cases (PUC's)

PUC-1: Create Plant Model

The user opens the software and accesses the GUI, where they can drag and drop nodes onto a canvas to represent real components. Connections between nodes can be created using edges, establishing the relationships and workflows between different parts of the plant. Users can customize the properties of each node as defined in the system databases and save the configuration as a model.

PUC-2: Run Plant Simulation

After building a plant model, users can configure input parameters as defined in the system table as well as time periods for simulation. The system will use the existing Python models in the back end to perform computations and provide outputs in the GUI.

PUC-3: Save and Load Plant Models

Users can save the plant models into the database for later retrieval and modifications. Each saved model will include all node properties, connections, and configuration data, allowing users to resume their work or create different versions of the model.

PUC-4: View Simulation Results

The results of any simulation run are presented to the user in a relevant format within the GUI.

PUC-5: Export Data

The system provides functionality to export simulation results and configuration data to common formats like CSV or Excel. This allows the user to leverage other Business Intelligence solutions to make better use of the data.

PUC-6: Manage User Accounts

Users can create and manage individual accounts, which will allow for secure access to their saved plant models and simulation data. Multi-user authentication will ensure that data is protected and that different users can work on their own models independently.

9 Functional Requirements

9.1 Functional Requirements

FR1

Description: The product shall allow users to create and visually edit plant models by adding, deleting, or modifying nodes (representing machines) and edges (representing material streams).
Rationale: Users need the ability to design complex industrial plants using a graphical interface to ensure accuracy and ease of use during the modeling process.
Fit Criterion: A user can successfully add, delete, or modify nodes and edges in the plant model, and the changes are reflected in the graphical user interface.

FR2

Description: The product shall enable users to input variables and parameters for each machine and stream data for each edge before running simulations.
Rationale: Accurate simulations require users to define specific operating conditions for machines (e.g., temperature, pressure) and material streams (e.g., flow rates, composition).
Fit Criterion: Users are able to input variables for all nodes and edges, and the system successfully accepts and stores these inputs for use during simulations.

FR3

Description: The product shall integrate a custom solver to simulate the behavior of the plant model based on the input variables and parameters.
Rationale: The system must be able to perform simulations to generate outputs that help users analyze and optimize the plant's performance.
Fit Criterion: When the user runs a simulation, the product processes the model and input parameters, and produces a result based on the solver, which is viewable by the user.

FR4

Description: The product shall provide error messages when the model contains invalid or incomplete information (e.g., disconnected nodes or missing parameters).
Rationale: To ensure accuracy, the system must notify users of errors before running a simulation, allowing them to correct the issues.
Fit Criterion: When a user attempts to run a simulation with an incomplete or invalid model, the system displays a clear error message explaining the issue.

FR5

Description: The product shall allow users to save plant models locally in a standard file format.
Rationale: Users need the ability to save the current state of their model so that work can be resumed later or shared with others.
Fit Criterion: A user can save a plant model to a specified location, and the file contains all nodes, edges, and parameters for later use.

FR6

Description: The product shall allow users to load previously saved plant models from local storage.
Rationale: Users need to be able to reopen and edit plant models over multiple sessions.
Fit Criterion: A user can load a saved plant model from a file, and the application will restore all nodes, edges, and parameters as they were at the time of saving.

FR7

Description: The product shall support zooming and panning within the graphical user interface to allow users to view large or detailed plant models.
Rationale: Industrial plant models can become complex, and users need flexible navigation options to focus on different parts of the model.
Fit Criterion: Users can zoom in/out and pan across the model without losing visual fidelity or usability.

FR8

Description: The product shall allow users to export simulation results to a file format compatible with external analysis tools (e.g., CSV or Excel).
Rationale: Users may need to perform additional analysis on the simulation results outside the software.
Fit Criterion: After running a simulation, users can export the results to a CSV file for further analysis in tools like Excel or Python.

FR9

Description: The product shall validate input parameters before allowing the simulation to run.
Rationale: To avoid errors during the simulation, the system must ensure that all necessary input parameters are provided and valid.
Fit Criterion: If the user attempts to run a simulation with missing or invalid parameters, the system will display an error message and prevent the simulation from starting.

FR10

Description: The product shall provide real-time feedback as users modify the plant model, showing warnings if connections between nodes are missing or incorrect.
Rationale: This ensures that users are aware of potential issues as they build their model, preventing errors during simulation.
Fit Criterion: When a user creates or modifies nodes and edges, the system will highlight any invalid or incomplete connections in real-time.

10 Look and Feel Requirements

10.1 Appearance Requirements

LAFAR1

Description: The product must feature a clear and logically structured layout that organizes information hierarchically to reflect the plant's structure and process flow.
Rationale: A structured layout will help users understand complex systems, making it easier to navigate and interact with the software during the modeling process.
Fit Criterion: A user can successfully locate and navigate between different sections of the plant model without confusion, and information related to specific equipment and processes is presented in a clear, hierarchical manner.

LAFAR2

Description: The product must use a consistent design across all elements, including fonts, colors, icons, and widgets, to ensure a cohesive and professional appearance.
Rationale: Consistency in the design will reduce cognitive load and facilitate quicker recognition and interaction with the software components, enhancing user experience.
Fit Criterion: All screens within the software should use the same font styles, color palette, iconography, and widget design, ensuring uniformity throughout the interface.

LAFAR3

Description: The software must provide drag-and-drop functionality, allowing users to add, delete, or modify elements such as reactor, source, separator etc.
Rationale: Drag-and-drop interactions simplify the process of constructing and modifying complex plant models, enhancing usability and reducing time required for model development.
Fit Criterion: A user can successfully add, delete, or rearrange plant elements using drag-and-drop actions, and the changes are accurately reflected in the plant model.

10.2 Style Requirements

LAFSR1

Description: The software must maintain sufficient contrast ratios between text, background colors, and interactive elements to ensure readability and accessibility.
Rationale: Maintaining proper contrast ratios improves readability, making the software accessible to users with visual impairments and ensuring compliance with accessibility standards.
Fit Criterion: All text and interactive elements must have a contrast ratio of at least 4.5:1 against their background colors as defined by WCAG (Web Content Accessibility Guidelines). The software must pass accessibility checks for color contrast.

LAFSR2

Description: The software must utilize a neutral color scheme (e.g., grays, whites) with professional accent colors such as blue and green, reserved for emphasis and highlighting specific elements.
Rationale: A neutral and professional color scheme maintains a clean and organized interface, minimizing visual distractions and allowing users to focus on the important aspects of the plant model.
Fit Criterion: All screens and interface components must use a base color palette of neutral colors, with accent colors only used for actionable elements, alerts, or key information. No more than three accent colors should be present at any time.

LAFSR3

Description: The software must incorporate clear visual cues, such as highlighting, shading, or outlines, to indicate interactivity and selectable elements.
Rationale: Clear visual cues help users identify interactive elements such as buttons, input fields, and links, improving overall usability and navigation within the interface.
Fit Criterion: All interactive elements, such as buttons and links, must change appearance (e.g., background color or border) when hovered over or selected, providing users with clear feedback on the element's interactivity status.

11 Usability and Humanity Requirements

11.1 Ease of Use Requirements

UHE1

Description: Each component of the service should be labeled with a descriptive tag that indicates what that component is used for
Rationale: Since this is a GUI service, its important that the user is aware what different components of the interface are used for and to accomplish this we need clear labels
Fit Criterion: Every component should have a one to one relationship with a clear label so that each UI component is well documented

UHE2

Description: The UI control animations such as dragging and dropping nodes and edges should be smooth and without lag
Rationale: A significant component of the UI is the animations that go into moving nodes and edges onto canvas. Its important these animations don't have lag that interrupt the users actions and so these animations should be lagless
Fit Criterion: All animations should not buffer and should be verfied by third party reviews that the animations are smooth and seamless and easy to use

11.2 Personalization and Internationalization Requirements

UHPI1

Description: The system will use english everywhere throughout the system
Rationale: The system will be used by english language speaking users and so it makes sense that the system is built with the english language
Fit Criterion: Any text in the system will be written in english

11.3 Learning Requirements

UHL1

Description: There should be documentation that enables a new user to learn the system
Rationale: The system will be used by various new users and so it's important that new users are able to understand the system by following documentation
Fit Criterion: When the system undergoes user testing, users should be able to learn and understand the system by following documentation within half an hour

UHL2

Description: There systems controls should be intuitive and easy to understand such that a new user can easily understand how to operate it
Rationale: A schema designing software should take the complexity out of designing the schema to allow users to focus on design rather than operation of the software
Fit Criterion: When the system undergoes user testing, new users should have no more than one question regarding how the controls of the system should be used

11.4 Understandability and Politeness Requirements

UHU1

Description: The system will use language and terminology that can be understood by trained chemical engineers
Rationale: The systems primary users will be educated chemical engineers and so the lanuage should be capable of being understood by educated chemical engineers
Fit Criterion: Any chemical engineer with at least a bachelor's degree should be able to understand the terminology used in the system

11.5 Accessibility Requirements

UHA1

Description: The system can be used by any computer with a monitor, keyboard and mouse with all controls mapped to the keyboard and mouse
Rationale: The system should be accessible by a standard engineer with a PC and so its important that all thats needed to operate it are basic computer peripherals like a monitory keyboard and mouse.
Fit Criterion: All of the systems outputs should be displayed by a monitor and all inputs can be handled by a keyboard and mouse.

12 Performance Requirements

12.1 Speed and Latency Requirements

PS1

Description: User inputs/interactions into the system should have a less than 60 ms latency
Rationale: User inputs includ adding, moving and deleting nodes/edges from the canvas, these inputs should be quick and seamless so that the user can work without significant interruption and to make sure the system operates smoothly.
Fit Criterion: All user interactions/inputs will have less than 60 ms latency from input to GUI output

PS2

Description: Saving a schema to a filesystem should take less than 1 second
Rationale: Saving a schema should not take a needlessly long amount of time and should be fast enough where a user can save multiple times without annoyance
Fit Criterion: Saving a plant schema to the file system will have a maximum measured time of 1 second

PS3

Description: Restoring a schema from the filesystem should take less than 1 second
Rationale: Restoring a schema should not take a needlessly long amount of time and should be fast enough where a user can restore a schema painlessly
Fit Criterion: Restoring a plant schema from the file system will have a maximum measured time of 1 second

PS4

Description: Running the factory simulation model and getting outputs should take no longer than 10 seconds
Rationale: While running the factory system model might take some time due to it being an external system, its important that making a request to this model and getting an output should not take too long which is why we are capping the maximum time to 10 seconds
Fit Criterion: Running the model and getting an output will take a maximum measured time of 10 seconds.

12.2 Safety-Critical Requirements

This app does not require any safety-critical requirement since it's not a safety critical application.

12.3 Precision or Accuracy Requirements

PPA1

Description: All numerical user inputs will not be constrained or rounded by the system
Rationale: We do not want the system to apply its own assumptions on user inputs and so its important the system does not constrain user input and instead pass the numerical user input to the external processing model and let the model handle decisions regarding the input
Fit Criterion: The system will not round or otherwise modify/constrain numerical user inputs

PPA2

Description: All graphical user inputs (creating/editing deleting nodes and edges) that are shown via the GUI will be sent to the model as whats shown in the canvas
Rationale: Whatever the user sees on the canvas should match what the system sends to the model to avoid any misinterpretation between the system and the user
Fit Criterion: The exact model shown on the canvas will represent the data that is sent to the processing model

12.4 Robustness or Fault-Tolerance Requirements

PRF1

Description: All failures or errors that the model produces will be displayed to the user with information regarding the error or failure
Rationale: Errors and failures produced by the external processing model should not be hidden from the user, instead this information should be displayed to the user so that they can correct their inputs and try again
Fit Criterion: If an error or failure is produced by the external model, a context box opens displaying information about the error/failure to the user

PRF2

Description: All failures and errors from the system should be shown to the user with information regarding the error/failure
Rationale: Errors and failures produced by the system should be shown to the user so that they can either fix their input or submit a bug request to get the problem fixed
Fit Criterion: If the system results in an error or failure, a context box opens where the user is provided information regarding the error/failure

12.5 Capacity Requirements

PC1

Description: The system should be able to handle schemas with utpo 100 nodes and 1000 edges
Rationale: From the sample data, it appears that most schemas will have at most 100 nodes and 1000 edges so we can assume that this is the capacity limit for our schema
Fit Criterion: The system will be constrained to handle 100 nodes and 1000 edges

12.6 Scalability or Extensibility Requirements

PSE1

Description: The system should only need to be scalable to handle a single user
Rationale: Since this system won't be used by multiple users at once, we only need the system to scale up to handle a single user
Fit Criterion: The system will be stress tested such that it can handle the inputs from a single user

12.7 Longevity Requirements

PL1

Description: The system should be maintainable without developer interference for at least 1 year
Rationale: Since we are passing the system off to the sponsor after completion, it's important that the system can be maintained without developer support for a reasonable amount of time
Fit Criterion: The system will be maintainable without developer support for at least 1 year

13 Operational and Environmental Requirements

13.1 Expected Physical Environment

OEEP1

Description: The product shall operate on standard web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, on both desktop and mobile devices.
Rationale: Users will likely access the web application from various browsers on different devices, so cross-browser compatibility is essential.
Fit Criterion: The web application functions seamlessly on the latest versions of Google Chrome, Firefox, Edge, and Safari, without browser-specific issues.

OEEP2

Description: The product shall be optimized for devices with a minimum of 4GB of RAM, and it shall function effectively on devices with at least a dual-core processor.
Rationale: Web applications are resource-light compared to desktop applications, but plant simulations may still be intensive.
Fit Criterion: The web application performs its core functions, including modeling and simulation, without excessive delays on devices meeting the minimum hardware requirements.

OEEP3

Description: The product shall provide a responsive interface, supporting screen resolutions from 1280x720 to 4K across different devices, including desktops, laptops, tablets, and smartphones.
Rationale: The web application needs to be accessible from a wide variety of devices and screen sizes.
Fit Criterion: The interface adapts to varying screen resolutions, providing a seamless user experience on both desktop and mobile devices without compromising usability.

13.2 Wider Environment Requirements

OEWE1

Description: The product shall require a continuous internet connection to perform all modeling, simulation, and collaboration tasks.
Rationale: The web application is designed to work online, and all features, including model creation, simulation, and saving, require an active internet connection to function.
Fit Criterion: The web application fails gracefully if the internet connection is lost, with appropriate error messages prompting users to reconnect.

OEWE2

Description: The product shall use a local MongoDB database hosted on the user's server or local machine for all data storage.
Rationale: Users will maintain a local instance of MongoDB, ensuring that all data, including models and simulations, are stored and managed locally without reliance on external cloud services.
Fit Criterion: The web application successfully connects to the local MongoDB instance for data storage and retrieval, with no external cloud dependencies.

13.3 Requirements for Interfacing with Adjacent Systems

OEIA1

Description: The product shall allow users to export and import data using common formats such as CSV, Excel, and JSON to integrate with adjacent systems.
Rationale: Users need to transfer data between the web application and external tools for advanced analysis and reporting.
Fit Criterion: The web application allows users to export and import simulation data, model configurations, and results in CSV, Excel, or JSON formats.

OEIA2

Description: The product shall support API integration, enabling seamless data exchange with third-party tools such as simulation engines and enterprise resource planning (ERP) systems.
Rationale: Industrial users often work with various tools, so the web application must provide APIs to interact with other systems.
Fit Criterion: The application exposes RESTful APIs for data input and output, and it can communicate with external simulation engines or ERP systems without manual data transfers.

13.4 Productization Requirements

OEP1

Description: The product shall be distributed as a local executable that can be installed and run directly on a user's device without the need for additional server configurations.
Rationale: The product is intended to be used as a standalone application, accessible directly from the user's system without reliance on cloud infrastructure.
Fit Criterion: The product is available as an installer or a self-contained executable that users can download, install, and run on supported operating systems.

13.5 Release Requirements

OER1

Description: The product shall be released with full documentation, including an online help center, API documentation, and user guides.
Rationale: Comprehensive documentation is essential to assist users and developers in understanding and using the web application effectively.
Fit Criterion: The release includes an online help center, complete API documentation, and user guides accessible within the web application.

14 Maintainability and Support Requirements

14.1 Maintenance Requirements

MSM1

Description: All system maintenance must be completed at least one week prior to the release or any critical deployment.
Rationale: The product is being developed with key deadlines, and no updates should occur within one week of a major release to ensure a stable version for users. This reduces the risk of introducing bugs right before the system goes live.
Fit Criterion: No system updates, patches, or changes are made within one week of a critical release or deployment.

MSM2

Description: Maintenance tasks shall be logged and tracked using an issue management system such as GitHub Issues.
Rationale: Proper tracking of maintenance ensures that all issues are documented, prioritized, and resolved efficiently, reducing potential downtime and improving software quality.
Fit Criterion: All maintenance issues are logged, with at least 90% of maintenance tasks tracked in the issue management system, and their status is visible to developers and users.

14.2 Supportability Requirements

MSS1

Description: Comprehensive technical documentation shall be provided, covering installation, configuration, usage, troubleshooting, and best practices for using the web application.
Rationale: Since the product may not be actively maintained after its release, clear and detailed documentation is essential for new users to be able to install, configure, and operate the system independently.
Fit Criterion: At least 80% of users should report in a post-release survey that they can operate the system solely based on the provided documentation.

14.3 Adaptability Requirements

MSA1

Description: The system shall be modular, allowing future developers to add new features or make changes to the existing codebase with minimal effort.
Rationale: Although the initial product release may not require adaptability, future development may involve extending functionality or adapting to new requirements, so the codebase must support modularity and flexibility.
Fit Criterion: New modules or features can be integrated into the system without requiring major changes to the existing codebase, as verified by a developer inspection report.

MSA2

Description: The system shall be built using industry-standard tools and frameworks to ensure future developers can easily adapt and work on the project.
Rationale: Using widely recognized frameworks and tools will make the system more maintainable and adaptable, as future developers will not have to learn proprietary or uncommon technologies.
Fit Criterion: The system uses well-known and widely adopted technologies such as React, Node.js, and MongoDB, ensuring that new developers can quickly familiarize themselves with the project.

15 Security Requirements

15.1 Access Requirements

SEC-ACC1

Description: ~~The system shall enforce role-based access control (RBAC) to restrict access to specific features and data based on user roles.~~
Rationale: ~~RBAC ensures that users can only access information and perform actions that are relevant to their roles, minimizing security risks associated with excessive permissions.~~
Fit Criterion: ~~The system defines roles such as Administrator, Operator, and Viewer, each with a distinct set of permissions and access rights.~~
Explanation: This requirement is being removed since our sponsor no longer requires us to have role based access control and so we do not need to consider this as a requirement within the scope of this project.

15.2 Integrity Requirements

SEC-INT1

Description: The system shall ensure data integrity by validating and sanitizing all input fields.
Rationale: Validating and sanitizing inputs prevent malicious data from being processed by the system, reducing the risk of attacks such as SQL injection or XSS.
Fit Criterion: All inputs are validated against predefined formats, and inputs containing scripts or special characters are automatically sanitized or rejected.

SEC-INT2

Description: The system shall implement version control for configuration files and database entries to prevent unauthorized or untracked modifications.
Rationale: Version control helps maintain the integrity of configuration files and critical data by allowing the system to track changes and revert to previous states if needed.
Fit Criterion: The system uses a version control system (e.g., Git) for configuration files, and database entries are tracked using timestamps and unique identifiers.

15.3 Privacy Requirements

SEC-PRI1

Description: The system shall implement data encryption for sensitive user information both in transit and at rest.
Rationale: Encrypting sensitive data protects against unauthorized access and data breaches, ensuring compliance with privacy standards.
Fit Criterion: All sensitive data, such as user credentials and personal information, is encrypted using AES-256 at rest and SSL/TLS during transmission.

15.4 Audit Requirements

SEC-AUD1

Description: The system shall log all relevant user actions, including login attempts, data access, and modifications, for audit and monitoring purposes.
Rationale: Comprehensive logging helps in tracking user activities, detecting suspicious behavior, and conducting post-incident analysis.
Fit Criterion: The system records all user actions, including timestamps, user IDs, IP addresses, and the type of action, with logs stored securely and reviewed periodically.

15.5 Immunity Requirements

SEC-IMM1

Description: The system shall employ secure coding practices and automated security testing to detect and prevent vulnerabilities during development.
Rationale: Secure coding and testing reduce the likelihood of vulnerabilities being introduced into the system, enhancing its resilience against attacks.
Fit Criterion: The system undergoes automated security testing with tools like SAST (Static Application Security Testing) and DAST (Dynamic Application Security Testing) integrated into the development lifecycle.

16 Cultural Requirements

16.1 Cultural Requirements

CR1

Description: The product shall be developed with consideration for the cultural practices and expectations of its users in an industrial setting, particularly in chemical plant operations. The primary language of the interface shall be English, and all symbols, colors, and terminology used in the product must be appropriate for professional use in business environments.
Rationale: The product will be used in business settings like chemical factories, where clarity and cultural sensitivity are important to avoid misinterpretation.
Fit Criterion: The system will undergo a review for every release to ensure that all symbols, colors, and terms are appropriate for use in industrial environments.

17 Compliance Requirements

17.1 Legal Requirements

LR1

Description: The product shall comply with IP protection laws, with all IP developed (including code, database structures, and algorithms) being transferred to Dr. Mahalec under a proprietary license. The system must also adhere to data security laws to protect proprietary plant data and user information ensuring no data is leaked to unauthorized actor.
Rationale: It is crucial to ensure proper IP protection because the product will be used in business settings that involve sensitive chemical plant operations. Compliance with data security laws ensures the confidentiality of user data and plant-specific information.
Fit Criterion: A legal review will be conducted at the end of the project when Dr. Mahalec purchases the software IP. This review will confirm the transfer of IP under a proprietary license and ensure compliance with data security laws. The documentation will confirm that no unauthorized access to the system's information is possible, and this will be included in the final handover.

LR2

Description: The system shall not use any copyrighted or patented technology without proper licensing or permission from the owner. This applies to third-party libraries, APIs, or external software integrated into the product. Any technology used must be appropriately licensed for commercial use and included in the final delivery.
Rationale: The product will be using several third-party components during development. Ensuring that none of these components violate copyright or patent laws is important to prevent legal complications.
Fit Criterion: A review will be conducted by Dr. Malaec's legal team as well as by our capstone team to verify that no unlicensed or unauthorized technology (libraries or code) is used. All third-party software licenses will be documented, and necessary permissions will be secured.

17.2 Standards Compliance Requirements

Discussed with Dr. Mahalec and team, currently the product that we are developing doesn't need to be any specific industry or software standards compliant due to the internal nature of its use and the fact that it will be handed over for proprietary use within a controlled environment.

18 Open Issues

At this stage, the primary open issue concerns the selection of an Object-Relational Mapping (ORM) framework to effectively map data between the application and the databases. Given the architecture of our system, which involves both a SQL and NoSQL database, it's essential to choose an ORM that supports seamless integration with both database types or can be extended to support this functionality.

The decision on the ORM will affect several aspects of the project:

- **Consistency and Maintainability**: A robust ORM will ensure that data is consistently managed between the frontend and backend, reducing the risk of mismatches or data corruption.
- **Performance**: The ORM choice will directly influence how efficiently the system can handle database operations, especially with large data sets and complex queries.
- **Flexibility**: The ORM should provide enough flexibility to allow for future database schema changes without significant refactoring of the codebase.

To address this issue, the team is currently evaluating several ORM frameworks, including those that offer multi-database support (e.g., for both SQL and NoSQL). Research is ongoing to determine which framework best suits the specific requirements of our system, taking into account factors such as:

- Ease of use and integration
- Documentation and community support
- Performance with both types of databases
- Customization capabilities to handle specific data mapping needs

This decision is crucial as it will form the backbone of how data flows within the system, ensuring both scalability and reliability in future iterations of the product.

19 Off-the-Shelf Solutions

19.1 Ready-Made Products

Currently, there are no available products that can directly solve the unique challenges we are addressing. Our product is a custom solution aimed at modeling industrial plants using a node-and-edge-based system, combined with a simulation feature. Given the specificity of this domain and the custom simulation script provided by our sponsor, this will be the first of its kind.

However, while a complete solution does not exist, there are third-party tools that can assist in certain aspects of our project, particularly for the front-end and back-end development. These include:

- ReactFlow (for node and edge management)
- ExpressJS (for building the back-end)
- PostgreSQL and MongoDB (for data management)

19.2 Reusable Components

Though the core functionality must be developed from scratch, we can leverage several reusable components to streamline specific areas of the project:

- **ReactFlow:** This third-party library will manage the visual representation of nodes, edges, and the canvas in the front-end, making it easier to build and interact with the plant models.
- **PostgreSQL and MongoDB:** Widely used for managing system and user data, these databases will provide reliable, scalable data storage solutions.
- **Python Simulation Script:** Provided by our sponsor, this Python script will handle plant simulations, meaning no additional development is required for the core simulation logic.

19.3 Products That Can Be Copied

As our solution is designed to address a specific problem with no direct off-the-shelf equivalent, there are no complete products that can be copied or modified to meet our needs. However, we can draw inspiration from similar industrial modeling tools and best practices to inform the design of our system. Additionally, we will use and

customize open-source libraries, such as ReactFlow, to meet our product's unique requirements.

20 New Problems

20.1 Effects on the Current Environment

Rather than using Excel tables to interact with the python model, our system will enable the user to use a GUI to create the plant schemas which will then interact with the python model. This should be a seamless transition since the GUI will be simpler and more user friendly than the Excel tables and it should not cause any conflicts in the overall environment.

20.2 Effects on the Installed Systems

The system will interface with the python model by making a request to an endpoint that will then process the input data via the python model. The endpoint will then respond to this request with the appropriate output data.

20.3 Potential User Problems

Existing users will need to learn how to use the schema building tools. However this should not be too difficult as it just requires using a mouse and keyboard to drag, drop and modify nodes/edges which should be as simple or simpler than inputting the data manually through Excel.

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

We will need to customize our input data for the endpoint such that it fits with the specification of the Python model. This may require massaging and modifying the data so that the values stay consistent but the form fits the Python model which may be a difficult task. However we think that the changes to the shape of the data should not be too extreme and should be doable within the scope of the project.

20.5 Follow-Up Problems

We will need to figure out how to create an efficient local deployment of the overall service in a way thats easy to setup for the user. This local deployment should be capable of spinning up a database, hosting an API server and building the client.

21 Tasks

21.1 Project Planning

The project schedule will follow the deliverables deadlines outlined in the SFWRENG 4G06 course outline.

Task	**Date**
Hazard Analysis 0	October 23
V&V Plan Revision 0	November 1
Proof of Concept Demonstration	November 11 to November 22
Design Document Revision 0	January 15
Revision 0 Demonstration	February 3 to February 14
V&V Report Revision 0	March 7
Final Demonstration (Revision 1)	March 24 to March 30
EXPO Demonstration	April TBD
Final Documentation (Revision 1)	April 2

Table 4: Project Timeline

21.2 Planning of the Development Phases

The development process will be divided into three major phases: GUI Development, Database and Schema Design, and Integration with Python Models. These phases will run in sequence, though some aspects of testing and integration will overlap as needed to ensure consistency throughout the project.

21.2.1 Phase 1: GUI Development

The primary goal of this phase is to create an intuitive, interactive interface that allows users to visualize and manipulate plant models.

In this phase, the team will first familiarize themselves with React Flow, which will be used to implement the node and edge creation features. The drag-and-drop functionality will be developed to enable users to place and organize components (nodes) on a canvas and connect them with arcs (edges). This system will serve as the foundation for building plant schemas within the software. Testing will ensure that the interface responds smoothly to user inputs and interactions, verifying that the canvas accurately represents the plant's structure as users modify it.

21.2.2 Phase 2: Database and Schema Design

In this phase, the focus will be on designing a schema that stores network topology data, including nodes and arcs. This phase will involve setting up and structuring the databases, ensuring they are capable of storing user configurations, plant data, and model results.

The team will establish both SQL and NoSQL databases (PostgreSQL and MongoDB) to separate user-specific and system-specific data. Integrating these databases with the front-end will allow for real-time data flow between the user interface and the back-end, ensuring that changes made in the GUI are reflected in the database. This phase will also include rigorous testing to ensure data consistency and integrity as users interact with the interface.

21.2.3 Phase 3: Integration with Python Models

The final phase will focus on connecting the graphical interface with the existing Python simulation models provided by Dr. Mahalec. The Python models will simulate plant operations based on the input data provided through the GUI.

To achieve this, APIs will be developed to facilitate communication between the front-end and the Python models. Once integrated, the outputs of the simulations will be displayed in the GUI, providing users with visual feedback on the simulation results. Testing will ensure that data is correctly sent to and from the Python models, and that the simulations produce accurate outputs based on user input.

22 Migration to the New Product

22.1 Requirements for Migration to the New Product

The project team requires that all newly created databases and functionalities developed for the new system maintain backward compatibility with the existing models. This ensures that legacy functionalities continue to operate as expected without the need for significant modifications. Additionally, the new databases and functionalities should be designed with modularity and flexibility in mind, allowing them to be accessed, modified, and utilized independently of the graphical user interface (GUI) system. This enables external systems, scripts, or tools to interact with the database and leverage its capabilities, promoting greater interoperability and ease of integration for future development or enhancements.

22.2 Data That Has to be Modified or Translated for the New System

The current system heavily relies on a collection of unstructured Excel spreadsheets to store and manage crucial data. These spreadsheets contain not only the data and the variables but also mappings that support the execution of Python scripts used within the system. Due to the lack of structure and standardized data management practices, a significant amount of manual intervention is required to maintain and utilize these spreadsheets effectively. This manual processing poses challenges for efficiency, accuracy, and consistency.

To ensure a seamless transition to the new system, all critical data currently stored in these spreadsheets must be systematically restructured, translated, and integrated into the new database architecture. This will involve converting unstructured and disparate data into a cohesive format that aligns with the new system's data models. Furthermore, the migration process must preserve the integrity and functionality of existing mappings and models, enabling the new system to interact with and utilize this data in an automated, streamlined manner without requiring additional manual intervention.

23 Costs

The only expense for the project so far is the price paid for React Flow Pro which is the pro version of the React Flow library which gives us access to sample React Flow components that we can integrate into our service. React Flow is a library

that enables developers to build GUI that include node and edge diagram creation. The service costs $195 in order to access the pro components which was paid by Dr. Mahelec. The other cost is time which is the 8 months of design and implementation time that we will need to fulfill the capstone.

24 User Documentation and Training

24.1 User Documentation Requirements

The user documentation will provide important information for users interacting with the system. Since the system is closed source and will be used by a specific set of industry professionals, documentation will focus on the core functionalities. This includes:

- A basic "Getting Started" guide included in the README file, providing an overview of system setup and usage.
- Step-by-step instructions for creating and modifying plant schemas using the GUI.
- Guidelines for managing and accessing plant models and user configurations within the database.
- A troubleshooting guide to resolve common issues related to database interactions or GUI operations.

24.2 Training Requirements

No formal training documentation is required. As the product is closed source and intended for use by Dr. Mahalec and his team, the system will be straightforward for them to operate with the provided documentation.

25 Waiting Room

If time permits we can try adding a live collaboration feature to the service such that multiple users can work on editing the canvas simultaneously. This would allow several users to work collaboratively on a project which would help improve its overall useability. However this is a feature that is not within the current scope of the project

and is more of a bonus feature that can be completed if given extra time which is why it's in the waiting room.

26 Ideas for Solution

- The front-end of the product will be developed using React, a popular JavaScript library for building user interfaces. We plan to utilize ReactFlow, a third-party library, to manage the nodes, edges, and canvas interactions within the modeling environment.
- The back-end will be built using ExpressJS, a lightweight and efficient web framework for Node.js, which will handle server-side operations and API requests.
- For data storage, we will use PostgreSQL to manage system-related data and MongoDB to handle user-related data, ensuring scalability and flexibility in data management.
- The simulation engine will integrate a Python script provided by our sponsor, which will be responsible for calculating the simulated values based on user input and model configurations.

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning. Please answer the following questions:

1. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

   Madhu: I will need to learn how to write frontend react code using TypeScript, I will also need to learn how to use the react flow library to build diagram building software.

   Dhrumil: I need to learn how to manage two databases within the same system using a single ORM (Object-Relational Mapping) tool. This involves understanding how to handle both SQL and NoSQL databases in a unified way, ensuring that data flows smoothly between the GUI and the backend. Additionally, I need to learn TypeScript since our frontend will use it, and it is important for developing the React components that is in react flow.

   Kishor: I will be learning how to implement the databases locally and setting up the relations for the SQL database.

   Rasik: I will need to learn how to effectively translate ambiguous requirements into an effective GUI through consistent collaboration with stakeholders using react, react-flow, typescript and modelling tools

2. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

   Madhu: For learning TypeScript I can either try to follow virtual courses that help teach TypeScript which can be found on YouTube or try to reimplement

an existing JavaScript using TypeScript which allows for experitiental learning. I will be looking at online courses since I find that I learn best by following a course.

Dhrumil: To learn how to manage both SQL and NoSQL databases with a single ORM, I will look into official documentation of the db and the orm and do code-along activities that focus on implementing small scale project. I think that hands-on practice will help me better understand how to connect both types of databases within one system. For learning TypeScript, I will be using a YouTube tutorial, as I find video-based learning easy to follow. Code-along exercises will also help me pick up the language quickly.

Kishor: For database implementation, I plan to use online resources, including official PostgreSQL and MongoDB documentation, along with tutorials that walk through the setup process and relational mapping. I may also reach out to my professor or classmates for additional guidance. I prefer learning through documentation and hands-on experience as it gives me the flexibility to try different configurations.

Rasik: To effectively translate requirements into a functional system, I will deepen my understanding of industry best practices in UI development. I plan to leverage online resources, such as courses and professional forums, to enhance my knowledge of React and other relevant tools. I have a strong interest in collaborating with stakeholders to ensure that the team delivers solutions that precisely meet project requirements in an efficient and effective manner. I believe this skill is crucial, particularly for individuals aspiring to take on leadership roles in the future.